



ANALYSIS PHASE - IT PROJECT

A central missing-person application

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Introduction

The Analysis Phase of the LocalAlert project focuses on examining the current methods used to communicate missing-person information and determine how the proposed LocalAlert application can address the identified weaknesses. The purpose of this phase is to develop a detailed understanding of the information requirements, user needs, problems associated with the current system and the functionality required from the proposed system. The analysis provides a link between the planning activities completed in Phase 2 and the detailed system design that will be undertaken in the following phase, this is described by Valacich, George and Hoffer (2020).

The analysis of LocalAlert focuses on the way missing-person information is currently reported, communicated and responded to by families, police personnel and members of the community. Existing communication methods may involve social media platforms, WhatsApp, Facebook, and direct communication between individuals. Although these methods can assist in distributing information, they do not provide a single centralized system for creating, managing, verifying and geographically distributing missing-person alerts. The analysis therefore examines the limitations of these approaches and identifies the requirements for a more organised and location-based solution.

Information gathering will be used to obtain a better understanding of the needs of the intended users and the activities associated with missing-person reporting. The selected information-gathering methods will assist in identifying how information is currently communicated, what information is required when reporting a missing person and what difficulties users may experience when attempting to respond to an alert.

The existing system will then be analysed by examining the processes involved in creating and distributing missing-person information. Particularly attention will be given to the speed of communication, verification of information, geographical targeting, coordination between families and authorities, and the ability of community members to provide information about a sighting. The weakness identified from this analysis will be used to establish the need for the proposed LocalAlert system.

Overall, the Analysis Phase will provide a detailed understanding of the current missing-person communication process and establish the requirements and data structures needed for LocalAlert. The findings from this phase will form the foundation for the system architecture, database design, interface and program design that will be developed during the System Design Phase, this is according to Dennis, Wixon and Roth (2021).

3.1. Information Gathering Methodology

The information gathering process for the LocalAlert project is intended to obtain a better understanding of how missing-person information is currently communicated and to identify the requirements that should be incorporated into the proposed application. Information gathering is an important part of the analysis phase because it allows the developer to understand the needs of potential users, identify problems with existing methods and determine the information and functionality required by the proposed system.

The information gathered through the following methods will be used to analyse the current system, identify its weaknesses and establish the functional and non-functional requirements of LocalAlert, Kendall (2019).

3.1.1. Observation

The observation process considers how information is presented when a person is reported missing, how important details are communicated, how widely the information can be distributed and how members of the public can respond when they have information about the missing person.

Several characteristics of the existing process are important to the analysis. Missing-person information shared through social-media platforms or messaging applications may be presented in different ways. The information may depend on the person creating the post to provide sufficient details, and users receiving the information may not necessarily be located close to the area where the person went missing. There may also be no central process for updating everyone who received the original message when new information becomes available or when the missing person has been found.

The observation therefore focuses on the following areas:

Observation Area	What is Examined	Relevance to LocalAlert
Information provided	Name, photograph, physical description, location information and report-to contact information	Determines the information required when creating a missing-person profile and alert
Distribution method	How alerts are communicated to members of the public	Identifies the need for a centralised alert distribution process
Geographic reach	Whether information is targeted according to the location of the incident	Supports the requirements for location-based notification

Verification	How users know whether information is legitimate	Supports verified family profiles and PIN-protected reporting
Public response	How members of the public provide information about sightings	Supports the Spotted Reporting feature
Alert updates	How users are informed when circumstances change	Supports alert escalation and resolution notifications
Alert closure	How the public is informed that the missing person has been found	Supports the alert-resolution function

Findings from Observation:

The observation of existing communication methods indicates that they can provide a fast way of distributing information, by they are not specifically designed as a central missing-person management system. Information can be distributed across multiple platforms making it difficult to maintain a single, consistent source of information.

A further issue is the lack of automatic geographic targeting. A Facebook or WhatsApp post may be shared with people who are far away from the location of the incident while potentially failing to reach people who are physically close to the area. LocalAlert addresses this problem through location-based alert distribution.

The observation also highlights the importance of providing sufficient information where an alert is created. LocalAlert therefore proposes the Family Link functionality, where information about family members can be prepared before an emergency occurs. This allows an authorised family administrator to select an existing profile rather than manually entering all information during an emergency.

The observation further supports the need for a structured method of reporting sightings. LocalAlert will allow a community member who receives an alert to submit a Spotted report containing information such as their location, directions and a photograph.

3.1.2. Participatory Analysis

Participatory analysis involves actively interacting with and examining existing methods and systems from the perspective of a potential user. Rather than only observing how a system operates, the developers consider the actual steps that a user would need to perform when attempting to report or respond to a missing-person case.

For LocalAlert, participatory analysis focuses on the user experience associated with existing communication methods and on the processes that would be required during a missing-

person incident. The analysis considers the activities that a family member, community member or police official would need to perform to communicate information.

The following activities are considered during the participatory analysis:

1. Creating a missing-person notification: examining the type of information that would need to be provided before other people could understand the situation.
2. Distributing the notification: considering how information would reach family members, police personnel and members of the community.
3. Locating relevant users: examining the difficulty of reaching people who are geographically close to the location where the person went missing.
4. Providing sightings: considering what information a community member would provide after seeing the missing person.
5. Updating an alert: considering how new information should be communicated to people who have already received an alert.
6. Closing an alert: considering how users should be informed when the missing person has been found.

Participatory Findings:

The participatory analysis indicated that the missing-person reporting process requires several activities to take place quickly. In an emergency, repeatedly entering information searching through different communication platforms and manually contacting different groups of people can increase the complexity of the process.

LocalAlert therefore combines these activities into a central application. A family administrator will be able to select a previously stored family member profile and initiate a missing-person alert after confirming the action using a four-digit PIN/

The analysis also identifies the importance of separating the different responsibilities of users. A family administrator should be able to create and resolve an alert, while a community member should primarily receive alerts and access provide Spotted reports, Police personnel should be able to receive alerts and access information relevant to their role.

The participatory analysis consequently supports the following proposed processes:

Family Administrator -> Select verified family member -> Confirm with PIN -> Create alert -> Distribute alert

and:

Community Member -> Receive alert -> Observe possible sighting -> Submit Spotted report -> Family/Police receive report

This process provides a more structured approach that relying entirely on independent messages across multiple communication platforms.

3.1.3. Interviews

Interviews are an information-gathering method that can be used to obtain direct information from potential users and stakeholders. For LocalAlert, interviews are relevant because different categories of users have different requirements when dealing with missing-person information.

The intended stakeholder groups include:

- Family administrator, who may need to create and manage missing-person alerts.
- Family members, who may need to receive information about a missing relative.
- Community members, who may receive local alerts and submit information about a sighting.
- Police personnel, who may receive and manage missing-person information.

The interviews for the analysis should focus on the users' experiences, expectations and concerns rather than asking participants to provide technical solutions. Questions should therefore concentrate on how missing-person information is currently communicated, what information users consider important and what features they would expect from a centralised application.

Proposed Interview Questions

No.	Interview Questions	Information Required
1	How would you normally receive information if someone in your community went missing?	Existing communication methods
2	What information would you expect to see in a missing-person alert?	Required alert information
3	How important is it for an alert to reach people close to the location where the person went missing?	Requirement for location-based alerts
4	What problems can occur when missing-person information is shared through WhatsApp or social media?	Weaknesses of existing methods
5	Would storing family-member information before an emergency be useful? Why?	Family Link requirements
6	What would make you trust that a missing-person alert is genuine?	Verification and Security requirements
7	If you saw the missing person, what information would you want to provide through the application?	Spotted Reporting requirements
8	How should users be informed when a missing person has been found?	Alert resolution requirements

9	Would receiving alerts based on your location be useful?	Location -based requirements
10	What would make a missing-person application easy to use during an emergency	Usability requirements
11	What privacy concerns would you have about storing photographs and personal information?	Privacy and security requirements
12	Would supporting South African official languages improve the accessibility of the application?	Language and accessibility requirements

Interview Information to be Collected

The responses obtained from potential users will be grouped into several requirements categories. This will make it easier to translate the information collected during interviews into a system requirement.

Requirement Category	Information to Determine
Alert creation	Information required to create a missing-person alert
User management	Different user roles and their permissions
Family Link	Information that should be stored about family members
Verification	Methods for reducing false or accidental reports
Location	How alerts should be distributed according to geographical location
Notifications	Who should receive alerts and updates
Spotted reports	Information required when a community member reports a sighting
Alert escalation	How and when the search area should expand
Alert resolution	How missing-person case should be closed
Privacy	Information that users consider sensitive
Accessibility	Language and ease-of-use requirements

Interview Findings and Their Relationship to LocalAlert

The information gathered through interviews will be used to validate the requirements identified during the preliminary investigation. Particularly attention will be given to the needs of the different user groups because the application must support different activities without giving every user the same level of access.

The expected requirements identified through the information-gathering process are consistent with the requirements established during Phase 2. Family admins require a quick and controlled method of reporting a missing family member. Community members require a

simple method of receiving relevant alerts and submitting sightings. Police personnel require access to information that can assist in managing missing-person cases.

The interviews will also help identify concerns relating to privacy and security. LocalAlert will handle information such as names, photographs, physical descriptions, locations and identification documents. Consequently, access to information must be controlled according to user's role.

3.2. Analysis of Existing System

The current system process for reporting and communicating information about missing persons is largely informal and depends on communication between the missing person's family, people within their personal networks, and the wider community. In practice, WhatsApp and Facebook are the main platforms used to distribute information about missing persons. These platforms provide a convenient way of reaching a large number of people because users can share posts and messages with groups, friends and other members of the community.

However, the existing process does not provide a central system for storing, verifying, updating and managing information about a missing person. Information is generally communicated through messages or social media posts, meaning that the effectiveness of the process depends heavily on individuals sharing the information correctly and ensuring that important details remain available.

The existing process can be described as follows:

Step 1: Family becomes concerned about the person's whereabouts

The process begins when family members become concerned because a loved one has not returned home or cannot be contacted. The family first attempts to determine whether there is a reasonable explanation for the person's absence.

Step 2: Family contacts people who may know the person's whereabouts

The family communicates with people who may have information about the missing person's whereabouts. This may include friends, relatives, neighbours or wider community or relevant authorities using a structured missing-person record.

Step 3: Initial enquiries do not locate the person

If the people contacted do not know where the person is, the family may determine that the person needs to be reported as missing and that information should be distributed more widely.

Step 4: A WhatsApp or Facebook post is created

The family or another individual may create a WhatsApp message/post or Facebook post containing information about the missing person. The post may include the person's name, photograph, physical description, contact details and the last known location.

However, rather than being kept as an organized record, these details are typically included in the post's caption or accompanying text. Because it might be challenging to maintain the information when the content is shared frequently, this causes a significant flaw in the current procedure.

Moreover, the creator of the post might not be the original reporting or a relative of missing individual. As a result of the missing individual. As a result, recipients of the message might not necessarily know where the information came from right once.

Step 5: Information is distributed through WhatsApp and Facebook

After, the message is shared via Facebook posts, WhatsApp groups, individual contacts, and other social media platforms. The reshare feature on WhatsApp enables a message to be shared with many people and groups.

Facebook users can share an existing post with their own networks using a similar method.

This makes it possible for word to get out beyond the family's increasing the number of people who could identify the missing person.

Step 6: Community members reshare information

Community members who receive the information may forward or reshare it with other people and groups. As the information moves between different groups and social media networks, multiple copies or versions of the original post may circulate.

This is one of the major weaknesses of the current system. Because the information is contained within a post or caption, important details can be lost, shortened, changed or separated from the original context during the sharing process.

For example, information such as:

- the telephone number that should be contacted;
- the missing person's physical description;
- the person's last known location; and
- other identifying information

may become less visible, be omitted when information is copied, or changed when someone creates another version of the post.

Consequently, a person receiving a reshared post may not have exactly the same information that was provided in the original report.

Step 7: Community member may identify or locate the missing person

A community member who receives the information may recognise the missing person, know their whereabouts, or encounter someone matching the description. They may then communicate this information to the family through WhatsApp, Facebook, telephone or another communication method.

The existing process therefore depends on the community recognising the person and manually communicating the information back to the family.

There is no standardised Spotter Report through which the location, directions, photograph and other sighting information can be submitted and associated with original missing-person alert.

Step 8: Information may continue circulating

Another issue occurs when the missing person is found. The family may inform people that the person has been located, but previously shared WhatsApp and Facebook posts may continue circulating.

Because copies of the original post may have already been forwarded or reshared across different groups and networks, it may be difficult for the family to ensure that everyone who received the original information also receives updates.

This can result in people continuing to search or report information about a person who has already been found.

3.2.1. Current Missing-Person Reporting Process Flow

Family becomes concerned



Family contacts people who may know the person's whereabouts



Person is not located



WhatsApp and Facebook posts are created



Missing-person details are added to the post/caption



Community members reshare or forward the post



Information reaches wider community



Community members may recognize/locate the person



Information is communicated back to the family

The major characteristic of this process is that information is repeatedly copied and distributed rather than being maintained in one central record.

3.2.2. Information Flow in the Existing System

The information flow can be summarized as:

Family → WhatsApp/Facebook → Community groups → Other users/groups → Possible sighting → Family

The process has no central database containing the current status and details of the missing person. Instead, information is distributed across different messages and social media posts.

This creates a distinction between information distribution and information management. WhatsApp and Facebook can distribute information effectively, but they are not designed to specifically to manage a structured missing-person case from the initial report through to this resolution.

The lack of a central record also means that updates do not necessarily reach everyone who received the original alert.

3.2.3. Main Findings from the Existing System

The analysis identifies several important characteristics of the current process.

First, WhatsApp and Facebook are effective communication channels because they are already widely used by communities. Their sharing functionality allows missing-person information to reach people beyond the family's immediate contacts.

Second, the process is highly dependent on manual sharing. The family or another individual must create the post, select where to distribute it and rely on other users to continue sharing it.

Third, important information is vulnerable to being lost or changed. Since details such as contact numbers, descriptions, and last known location are commonly included in captions or accompanying text, they can be omitted, shortened or altered as information is copied or reshared.

Fourth, the original source of information may become unclear. A person who creates or reshares a WhatsApp or Facebook post may not be the original reporter or family member. Consequently, recipients may have difficulties determining who initially reported the missing person or whether the information is still accurate.

Fifth, multiple versions of the same missing person information can circulate simultaneously. This can result in some people having outdated information while others have more recent information.

Finally, there is no central alert status or structured resolution process. When a missing person is found, the family can communicate the outcome, but previously shared copies may continue circulating.

These findings provide the basis for the proposal LocalAlert system. Rather than eliminating existing communication platforms, LocalAlert is intended to introduce a central, structured and location-based missing-person alert system in which the important information is maintained within the application. External social-media platforms can still potentially be used to extend the reach of verified alerts, while LocalAlert remains the central source for the missing-person record.

This directly addresses the problem identified in the existing process through features such as Family Link, verified family-member profiles, PIN-protected reporting, location-based alerts, automatic alert escalation, Community Spotted Reporting and alert resolution.

3.3. Data Analysis

The information that LocalAlert must gather, process, store, and disseminate in order to support the missing-person alert procedure was determined through data analysis. The information needed by family admins, family members, police officers, and community members was taken into account in the analysis. Pressman and Maxim (2019) explain that the sensitivity nature of data pertaining to children and families, as well as the necessity of maintaining accurate and consistent details throughout the alert lifespan, were given special consideration.

LocalAlert will save the data in organized records, in contrast to the existing procedure, where information is frequently included in Facebook and WhatsApp captions and may be lost or altered when postings are reshared. This makes it possible for details like a missing person, alert, location, and Spotted Report to stay connected to the relevant record.

The major categories of data required by the proposed system are:

1. User data – information required to identify and authenticate registered users
2. Family group data – information identifying family groups and their administrators.
3. Family member data – information about individuals who have been pre-loaded
4. Verification data – information relating to identify-document submission for family member verification.
5. Missing alert data – information about the missing person, last known location, alert status and alert timing.
6. Location data – geographical information used to determine which users and police station should receive an alert.
7. Spotted Report data – information submitted by community members when they believe they have seen the missing person.
8. Police station data – information required to identify the relevant local and neighbouring police stations.
9. Notification/recipient data – information used to determine which users have received an alert and should receive subsequent updates.
10. Language preference data – the preferred language selected by each user.

The proposal identifies family profile information as an important part of LocalAlert because information such as the person's name, photograph, age and physical description.

3.3.1. Data integrity

Data integrity refers to maintain the accuracy, consistency and reliability of data throughout its lifecycle. Data integrity is particularly important for LocalAlert because inaccurate

information could affect the ability of the community or police to identify and locate a missing person.

The following measures will be used to maintain data integrity:

Data integrity requirement	Application in LocalAlert
Unique identification	Users, family groups, family members, alert and Spotted reports will have unique identifiers
Required fields	Important information such as user's name, email address and an alert's associated family member cannot be left blank.
Referential integrity	Foreign keys will ensure that records such as alerts and Spotted Reports remain associated with valid family members and users.
Input validation	Email addresses, telephone numbers, dates, PINs and other appropriate fields will be validated before being stored.
Role validation	The system will verify that a user has the appropriate role before performing restricted operations.
Verification	A family member profile must have the required identity-document information before it can be used for a missing-person alert.
Status control	Alerts will have controlled states such as Active, Still Missing and Resolved
Location validation	Location information will be validated before being used for geographical alert distribution.
PIN protected	The family administrator must confirm important actions using their PIN.
Controlled updates	Important alert information should be updated through LocalAlert rather than creating unrelated copies of the alert

Because the current WhatsApp and Facebook method can generate different versions of the same missing-person information, the usage of structured records is very crucial. Instead, LocalAlert keeps record, which lessens the chance that crucial information from the initial alarm.

3.3.2. Data Constraints

The proposed system is subject to several data and operational constraints.

1. User authentication constraint

Only registered and authenticated users should be able to access functions appropriate to their role. This prevents unauthorised users from managing family profiles or creating and resolving missing-person alerts.

2. Role-based access constraint

Different users have different responsibilities. A family administrator should be able to manage family information and create or resolve alerts, while community members primarily receive relevant alerts and submit Spotter Reports.

3. Family-member verification constraint

A family member should not be available for use in a missing-person alert until the required identity-document information has been uploaded for verification.

4. PIN constraint

Missing-person reports and alert resolution require confirmation using a four-digit PIN. This reduces the possibility of accidental activation or resolution.

5. Location constraint

The location of the user and the last known location of the missing person must be sufficiently accurate for the radius-based notification process to operate correctly.

6. Internet connectivity constraint

The first version of LocalAlert requires an active internet connection for receiving and sending alerts. Offline functionality is therefore outside the initial version.

7. Privacy constraint

LocalAlert handles sensitive information, particularly information relating to children. Only information necessary for the operation of an alert should be made available to community users.

8. Storage constraint

The project is an academic prototype and therefore is limited resources compared with a production-scale national platform. Storage and infrastructure must remain appropriate for the scope of the project.

9. Language constraint

The proposed system must support South Africa's 11 official languages through a selectable language preference.

10. Police integration constraint

The project does not include full integration with SAPS computer systems. The proposed police functionality is therefore limited to the functionality defined within LocalAlert.

3.4. Weakness of the current system

The analysis of the existing system identified several weaknesses in the current WhatsApp and Facebook based approach

3.4.1. Information can be lost during resharing

One of the most significant weaknesses is that important missing-person information is often contained within the caption or accompanying text of a WhatsApp or Facebook post. When users reshare or recreate the post, information such as cellphone numbers, physical description and last known location can be left out.

For example, an original post may contain a photograph, the person's description, two contact numbers and the last location where the person was seen. A person who creates a new post using the same photograph may only copy the person's name and one contact number. The second version therefore does not contain all the information from the original report.

3.4.2. Information can be changed

Information can also be changed as posts are copied and redistributed. A person's description or last known location may be changed unintentionally or intentionally. This creates a risk that people responding to the alert may be working with inaccurate information.

3.4.3. Original source can become unclear

A person who creates or reshares a WhatsApp or Facebook post may not be the original reporter or a member of the missing person's family. As the post moves between groups, recipients may therefore be unable to determine who originally provided the information.

This creates a source-control problem. The information may appear genuine because it has been widely shared, even though the recipients cannot easily establish its original source.

3.4.4. Multiple versions of an alert can exist

The same missing person may be represented by several WhatsApp messages or Facebook posts. Each version may contain slightly different information.

This means that there is no guaranteed single version of the information that everyone can refer to.

3.4.5. Distribution depends on users

The existing process depends on individuals forwarding or resharing information. If people do not forward the message, the alert may not reach people outside the original social network.

3.4.6. No automatic geographical targeting

WhatsApp, Facebook, Instagram and X (formerly Twitter) do not provide the specific LocalAlert mechanism in which an alert can initially target a defined geographical radius and subsequently expand according to predetermined rules.

3.4.7. Search-area expansion is manual

If the person is not found, the family and community must manually continue sharing the information. There is no automatic mechanism that expands the notification area from 800 m to 20 km after 40 minutes.

3.4.8. Sighting information is fragmented

A community member who believes they have seen the missing person may send a WhatsApp message, make a telephone call or respond through another communication channel. The sighting information is therefore not necessarily stored alongside the original missing-person report.

3.4.9 Found notification are difficult to control

Once a missing person has been found, the family can communicate the outcome, but copies of the original social media posts may continue circulating. There is no central mechanism that can update every recipient of the original alert.

3.4.10. Lack of central alert lifecycle

The existing process does not provide a structured lifecycle showing whether an alert is active, has escalated, is still missing or has been resolved.

3.5. Analysis of the proposed system

The proposed LocalAlert system is designed to address the weaknesses identified in the existing process. The system will provide a central platform through which family admins can manage family-member information, create verified missing-person alerts and receive structured information from the community.

The proposed system will use a role-based approach involving:

- Family Administrators
- Family Members
- Community Members
- Police Personnel

The core process begins with a family administrator selecting a verified family member from the Family Link and activating a missing-person alert using PIN confirmation. The alert is then distributed according to the defined location rules. The proposal specifies that the local police station and community within an 800 m radius should receive the initial alert, with escalation to 20 km after 40 minutes if the person has not been found.

Community member who receives the alert can submit Spotter Report containing at least one useful piece of information, such as their live location, written directions or a photograph.

3.5.1. Functional requirements

The functional requirements describe that the proposed LocalAlert system must be able to do.

No.	Functional requirement
FR1	The system shall allow users to register using their full name, email address, telephone number, password, user role.
FR2	The system shall allow registered users to log in and log out securely
FR3	The system shall authenticate users before allowing access to protected functionality.
FR4	The system shall allow a family administrator to create a Family Link group
FR5	The system shall allow a family administrator to add family members to the family groups
FR6	The system shall allow family-member information such as full name, date of birth, gender, physical description and photograph to be stored
FR7	The system shall require the necessary identity-document information before a family member can be used in a missing-person report.
FR8	The system shall allow a family administrator to select a verified family member and initiate a missing-person report.

FR9	The system shall require a four-digit PIN confirmation before a missing-person alert is created.
FR10	The system shall create an alert using the selected family member's stored information.
FR11	The system shall record the last known location of the missing person.
FR12	The system shall notify linked family members when an alert is created.
FR13	The system shall notify the relevant local police station.
FR14	The system shall notify community users within an 800 m radius of the relevant location.
FR15	The system shall automatically expand the radius from 800 m to 20km
FR16	The system shall notify linked family members when the alert is escalated
FR17	The system shall notify linked family members when the alert radius is expanded.
FR18	The system shall allow community members who receive an alert to submit a Spotted Report.
FR19	A Spotted Report shall allow the community member to provide at least one of a live location, written directions or a photograph.
FR20	The system shall associate each Spotted Report with the relevant missing-person alert.
FR21	The system shall send Spotted Reports to the relevant family members and police personnel
FR22	The system shall allow the family administrator to resolve a missing-person alert using the four-digit PIN.
FR23	The system shall notify the original alert recipient when the missing person has been found.
FR24	The system shall move unresolved alerts to the Still Missing category after five days.
FR25	The system continues accepting Spotted Reports for Still Missing cases
FR26	The system shall allow a family administrator to resolve an alert
FR27	The system shall securely store uploaded files and restrict access to ID documents to authorised family admins and verified police users
FR28	The system shall display a privacy notice during registration and obtain consent for data collection in accordance with the project's POPIA requirement
FR29	The system shall support all 11 South Africa official language
FR30	The system shall allow users to select their preferred language.

3.6. Non-functional requirements

No.	Non-functional requirement
NFR1	The main screen should load within 3 seconds under stable conditions
NFR2	Norma API requests should respond within 2 seconds
NFR3	Passwords must be securely hashed
NFR4	Family-admin PINs must not be stored in plain text
NFR5	API requests must use JWT authentication.
NFR6	Invalid authentication tokens must be rejected
NFR7	Role-based access control must be implemented
NFR8	The application must support Android and iOS.
NFR9	Version 1.0 requires an active internet connection
NFR10	ID documents must only be accessible to authorised users
NFR11	The codebase must be structured and maintainable
NFR12	The system must follow POPIA principles.
NFR13	The system should collect only information necessary for its purpose
NFR14	Communication between the application and API must use HTTPS
NFR15	GPS-radius filtering must use appropriate geographic calculations
NFR16	The interface should be simple enough for emergency use
NFR17	The application should provide appropriate accessibility
NFR18	The application should provide consistent information across support devices.
NFR19	The system should prevent unauthorized modifications of alerts.
NFR20	The system should maintain reliable alert status information

3.7. Data modelling of proposed system

Data modelling is used to describe how information will be structured and how different data object within LocalAlert will relate to one another. A relational database model is appropriate because LocalAlert contains clearly defined relationships between users, family groups, family members, alerts, police station and Spotter Reports.

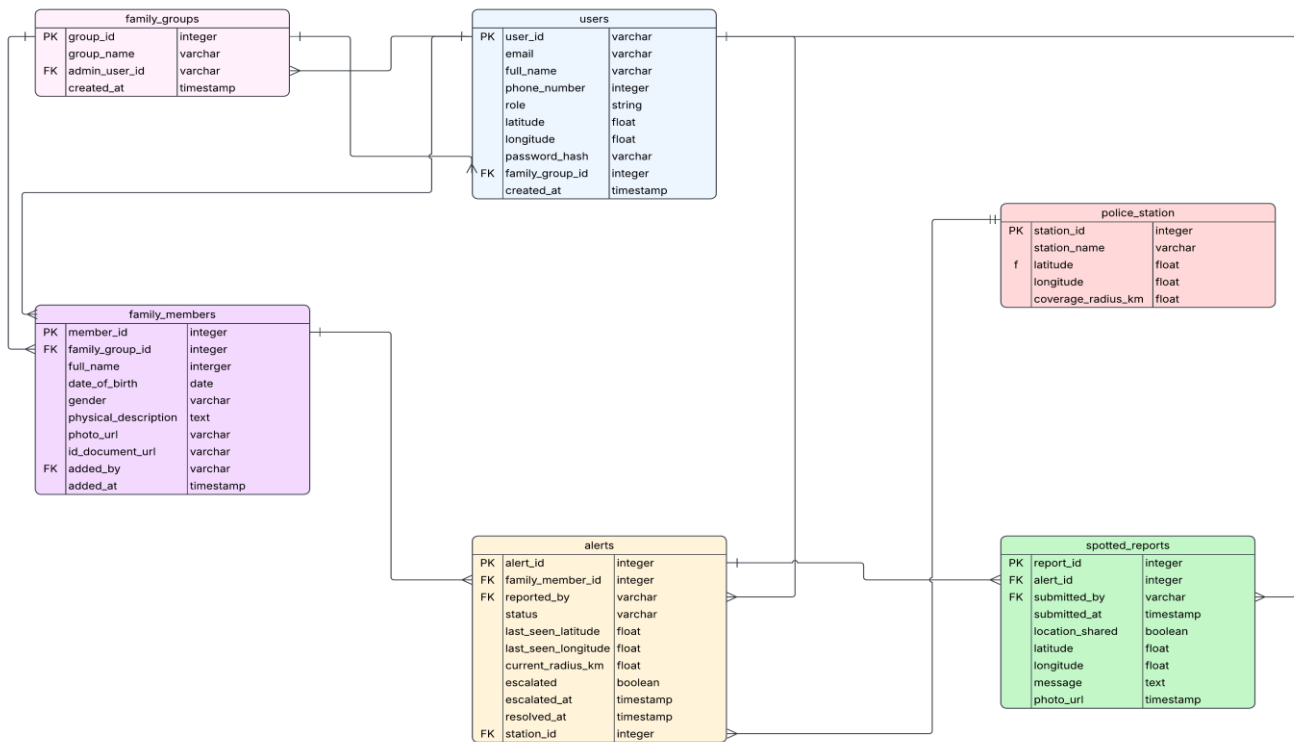
The proposed database will use PostgreSQL as the relational database management system. The database structure will apply primary keys foreign key, appropriate data types and constraints to maintain data integrity. Coronel and Morris (2019) identify entity-relationship modelling and normalization as important parts of database design.

The principal entities are:

1. users
2. family_groups
3. family_members
4. alerts
5. spotted_reports
6. ploce_station

The model separates information into related tables rather than placing all information into one large table. This reduces duplication and makes it easier to update information consistently.

3.7.1. Entity Relationship Diagram



3.7.2. Data Dictionary

A data dictionary defines the fields contained in the proposed database and provides a common description of the information stored in the system. Data dictionaries are used during systems analysis to describe the structure of data flows and stored data (Kendall, 2019)

users Table

Field	Data Type	Description	Constraints
user_id	UUID	Unique identifier for the user	PK, NOT NULL
full_name	VARCHAR(100)	User's full name	NOT NULL
email	VARCHAR(150)	User's email address	UNIQUE, NOT NULL
phone	VARCHAR(20)	User's telephone number	NOT NULL
password_hash	VARCHAR(255)	Securely hashed password	NOT NULL
role	VARCHAR(30)	User role	NOT NULL
language_preference	VARCHAR(50)	Selected language	NOT NULL
created_at	TIMESTAMPS	Account creation date/time	NOT NULL

family_groups Table

Field	Data Type	Description	Constraints
group_id	UUID	Unique family-group identifier	PK, NOT NULL
admin_user_id	UUID	Family administrator	FK -> USERS
group_name	VARCHAR(100)	Name of the family group	NOT NULL
created_at	TIMESTAMPS	Data/time group was created	NOT NULL

family_members Table

Field	Data Type	Description	Constraints
member_id	UUID	Unique family member identifier	PK, NOT NULL
group_id	UUID	Family group to which member belongs	FK, NOT NULL
member_name	VARCHAR(100)	Full name of family member	NOT NULL
date_of_birth	DATE	Date of birth	NOT NULL
gender	VARCHAR(30)	Gender	NOT NULL
physical_description	TEXT	Physical description	NOT NULL
profile_photo	VARCHAR(255)	Location/reference of photograph	NOT NULL
id_document	VARCHAR(255)	Identity-document reference	NOT NULL
verification_status	VARCHAR(30)	Verification state	NOT NULL
created_at	TIMESTAMP	Date/time profile as created	NOT NULL

alerts Table

Field	Data Type	Description	Constraints
alert_id	UUID	Unique alert identifier	PK, NOT NULL
member_id	UUID	Missing family member	FK, NOT NULL
reporter_id	UUID	User who created alert	FK, NOT NULL
last_known_location	TEXT	Description of last known location	NOT NULL
latitude	DECIMAL(10,7)	Latitude of last known location	NOT NULL

longitude	DECIMAL(10,7)	Longitude of last known location	NOT NULL
status	VARCHAR(30)	Alert lifecycle status	NOT NULL
radius_km	DECIMAL(8,2)	Current notification radius	NOT NULL
created_at	TIMESTAMP	Alert creation time	NOT NULL
escalated_at	TIMESTAMP	Time of escalation	NULL
resolved_at	TIMESTAMP	Time alert was resolved	NULL

spotted_reports Table

Field	Data Type	Description	Constraints
report_id	UUID	Unique Spotted Report identifier	PK, NOT NULL
alert_id	UUID	Associated missing-person alert	FK, NOT NULL
reporter_id	UUID	User submitting report	FK, NOT NULL
latitude	DECIMAL(10,7)	Report location latitude	NULL
longitude	DECIMAL(10,7)	Report location longitude	NULL
directions	VARCHAR(255)	Written directions/description/live location	NULL
photo	VARCHAR(255)	Photograph reference	NULL
created_at	TIMESTAMP	Time report was submitted	NOT NULL

police_stations Table

Field	Data Type	Description	Constraints
station_id	UUID	Unique police station identifier	PK, NOT NULL
station_name	VARCHAR(150)	Police station name	NOT NULL
address	TEXT	Physical address	NOT NULL
latitude	DECIMAL(10,7)	Station latitude	NOT NULL
longitude	DECIMAL(10,7)	Station longitude	NOT NULL
contact_number	VARCHAR(20)	Station contact number	NOT NULL

alert_recipients Table

Field	Data Type	Description	Constraints
recipient_id	UUID	Unique notification recipient record	PK, NOT NULL

alert_id	UUID	Alert that was distributed	FK, NOT NULL
user_id	UUID	User who received the alert	FK, NOT NULL
notified_at	TIMESTAMP	Time notification was sent	NOT NULL
notification_radius	DECIMAL(8,2)	Radius used when recipient was notified	NOT NULL

3.7.3. Database Schema

PostgreSQL will be used as the relational database management system for LocalAlert. This is consistent with the Phase 2 technology selection.

The logical database schema is:

USERS

PK user_id

email

full_name

phone_number

role

latitude

longitude

password_hash

language_preference

created_at

FAMILY_GROUPS

PK group_id

FK admin_user_id → USERS.user_id

group_name

created_at

FAMILY_MEMBERS

PK member_id

FK family_group_id → FAMILY_GROUPS.group_id

full_name

date_of_birth

gender

physical_description

photo_url

id_document_url

verification_status

added_at

POLICE_STATION

PK station_id

station_name

latitude

longitude

coverage_radius_km

ALERTS

PK alert_id

FK family_member_id → FAMILY_MEMBERS.member_id

FK reporter_id → USERS.user_id

FK station_id → POLICE_STATION.station_id

status

last_seen_latitude

last_seen_longitude

current_radius_km

escalated

escalated_at

resolved_at

created_at

SPOTTED_REPORTS

PK report_id

FK alert_id → ALERTS.alert_id

FK submitted_by → USERS.user_id

submitted_at

location_shared

latitude

longitude

directions

photo_url

ALERT_RECIPIENTS

PK recipient_id

FK alert_id → ALERTS.alert_id

FK user_id → USERS.user_id

notified_at

notification_radius

Normalisation

Database normalisation is used to organise relational data and reduce unnecessary redundancy. Elmasri and Navathe (2016) and Coronel and Morris (2019) explain that normalisation helps reduce insertion, update and deletion anomalies.

First Normal Form (1NF)

The LocalAlert tables satisfy the principle of 1NF because each field stores an atomic value rather than a repeating group. For example, the USERS table stores one email address per record rather than a list of email addresses.

Second Normal Form (2NF)

The design uses single-column primary keys, meaning there are no partial dependencies on only part of a composite primary key. Non-key attributes depend on the relevant entity's primary key.

Third Normal Form (3NF)

The design separates information that belongs to different entities. For example, police-station information is stored in POLICE_STATION rather than repeatedly storing the station name and coordinates in every alert. Similarly, family-group information is stored in FAMILY_GROUPS.

This structure reduces duplicated data and makes updates more reliable (Hoffer, Venkataraman and Topi, 2019).

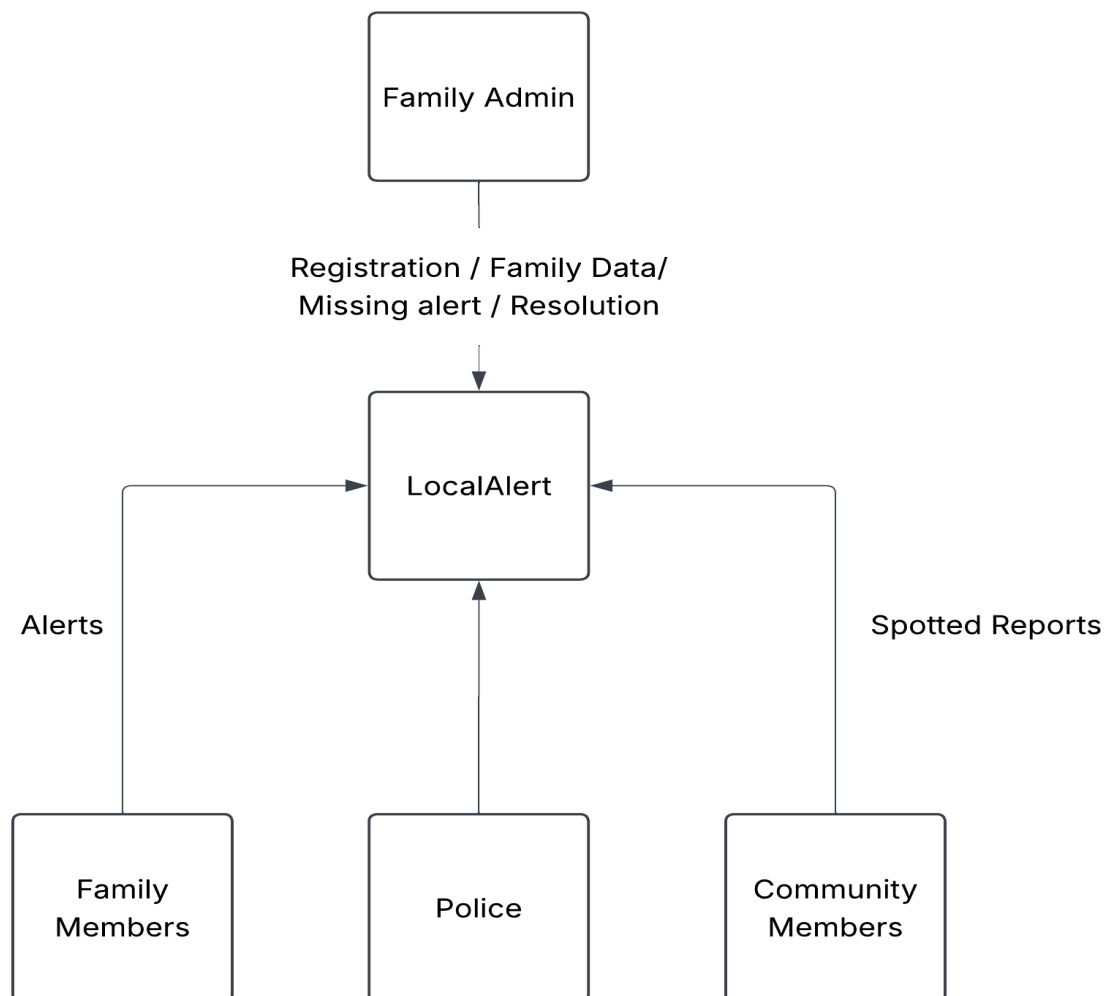
3.7.4. Data Flow Diagram

A Data Flow Diagram represents how data moves between external entities, processes and data stores. DFDs are particularly useful during systems analysis because they describe what the system does without unnecessarily focusing on implementation details (Kendall, 2019)

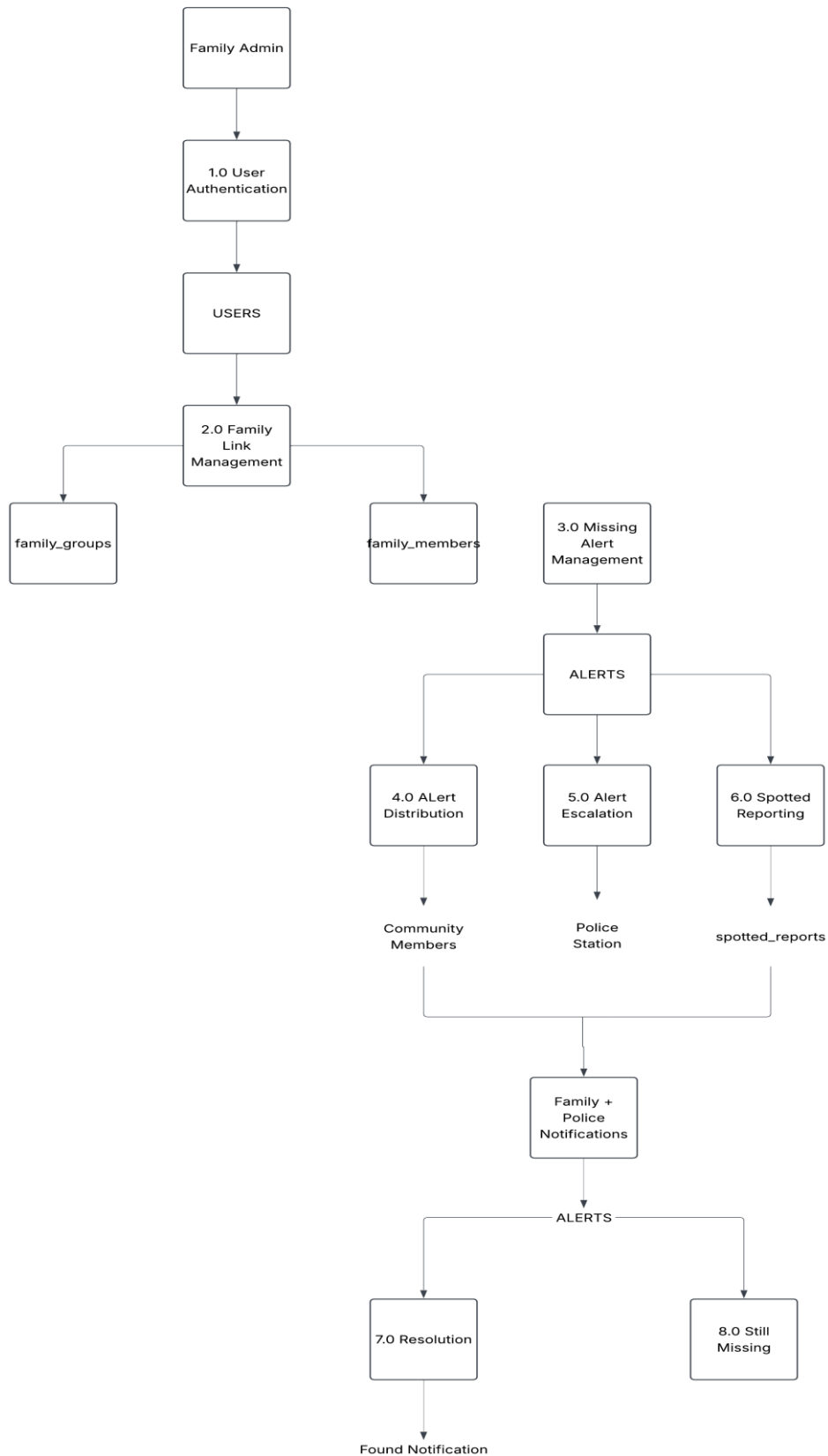
The LocalAlert DFD should use four basic components:

Symbol	Meaning
Rectangle	Person or external system interacting with LocalAlert
Process	Activity that transforms incoming data
Data Store	Stored information
Arrow	Movement of information

Level 0 Context Diagram: High-level view of LocalAlert as one system



Level 1 Context Diagram: Decomposition of the LocalAlert system into its major processes



1.0 User Authentication

This Process handles:

- Registration
- Login
- Logout
- Role identification
- JWT authentication

This process communicates with the users data store

2.0 Family Link Management

This process manages:

- Family groups
- Family administrators
- Family member profiles
- Verification information
- Profile information
- Profile photographs
- ID-document files

It communicates with:

- users
- family_groups
- family_members
- Secure file storage

3.0 Missing Alert Management

The family administrator selects a verified family member and enters the required PIN. The system validates the request and creates an alert.

This process interacts with:

- family_members
- alerts
- alert_recipients
- Secure file storage

4.0 Location-Based Alert Distribution

The system uses the alert location and current search radius to determine the appropriate recipients.

The initial radius is 800 m, as specified by Phase 2

This process uses:

- Active alert location
- User location
- Police station location
- Notification radius

The Haversine formula specified in Phase 2 NFRs can be used to calculate geographical distance.

5.0 Alert Escalation

If the person has not been found after 40 minutes, the system changes the search radius from 800 m to 20 km.

6.0 Spotted Reporting

A community member who receives an alert can submit a sighting

The report can contain:

- Live GPS location
- Written directions
- Photograph

The system associates the report with the relevant alert and sends it to the family and police.

7.0 Alert Resolution

After the missing person has been found, the family administrator enters the four-digit PIN.

The system changes the alert to a resolved state and sends the found notification to the original recipients.

8.0 Still Missing Management

If an alert remains unresolved for five days, the system moves it into the Still Missing category

The alert remains available for Spotted Reports. This directly implements the Phase 2 requirements

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